Week :8 (19-23 October 2026)

من – إلى الأسبوع #:

Subject/الموضوع: Physics **Unit Title/الوحدة عنوان: Wave phenomena** **Grade/: 12**

Lessons الحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formative تكويني/ Summative ختامي / Differentiation التمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1: الحصة 1 قانون بويل	I can explain and apply the principles of **diffraction and interference** to describe the behavior of waves.	2.2 Clarify syntax/structure and highlight relationships in text, formulas, diagrams. 2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	please check student Teams
Lesson 2 الحصة الثانية	I can explain **Young's double-slit experiment** and use the interference pattern to determine wavelength, slit separation, or fringe spacing.	1.1 Flexible display of information (size, color, font, layout). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports.	please check student Teams
Lesson 3 الحصة الثالثة	I can conduct my **Physics IA experiment systematically and safely**, collecting sufficient high-quality data to investigate my research question.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 2.1 Clarify vocabulary and symbols using visuals, connections, and supports. 3.2 Highlight big ideas, patterns, and critical features using organizers and examples.	please check student Teams
Teacher's notes/ Reminders ملاحظات المعلم والتذكير	Bring a charged devices, calculator, Physics notebook, ruler, pencil and coloured pens daily. Complete all graph and calculation work, participate actively, review feedback and upload required work to Teams.		

#الأسبوع Week # 4

Technology Integration دمج التكنولوجيا	Digital graphing or teacher-provided graph displays; calculators for gradients, areas and equations; motion videos or simulations where available; Teams for resources, feedback and submission.
Cross-curricular links الروابط بين المواد الدراسية	Mathematics: gradient, geometric area, algebra, rearranging equations, proportional reasoning and negative values. English: interpret motion descriptions and justify conclusions. UAE engineering and road safety: analyse transport data and stopping motion.
External Assessment Integration دمج التقييمات الخارجية	NGSS-style external assessment questions requiring students to analyse data, interpret multi-section graphs, calculate gradient and signed area, select kinematics equations, translate between representations and justify conclusions.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	ATL Thinking: analyse patterns, connect representations and evaluate solutions. Communication: explain graph features and justify equation choices. Self-management: organise multi-step working and check signs and units. Collaboration: critique reasoning respectfully. Wellbeing: build confidence through scaffolded practice and purposeful challenge.
Standards المعايير	HS-PS2-1: Analyze data to support the claim that Newtons second law of motion describes the mathematical relationship among the net force on a macroscopic object, its mass, and its acceleration. Supporting alignment: students analyse motion data and graphical acceleration as the kinematic foundation for later force-motion modelling.
Inquiry/ TOK Question البحث والاستقصاء أسئلةTOK	How can the same journey be represented by equations and different motion graphs, and how can we verify that the representations are consistent?
Success Criteria معايير النجاح **(Measurable)**	**Lesson 1** **Must: Explain how coherent waves from two slits produce alternating bright and dark fringes.** **Should: Apply the relationship \[s=\frac{\lambda D}{d} \]where \(s\) = fringe spacing, \(\lambda\) = wavelength, \(D\) = slit-to-screen distance, and \(d\) = slit separation.** **Could: Analyze experimental data from a double-slit pattern, calculate an unknown quantity, and evaluate sources of uncertainty.** **Lesson 2:** **Must: Describe diffraction and identify the conditions that produce significant diffraction.** **Should: Explain constructive and destructive interference using path difference and phase difference.** **Could: Apply diffraction and interference concepts to analyze unfamiliar wave situations and justify the observed pattern.** **Lesson3:** **Must: Set up and carry out my experimental method safely, identifying the independent, dependent, and controlled variables.** **Should: Collect and record sufficient quantitative data, including units and measurement uncertainties, in a well-organized table.** **Could: Evaluate the quality of my preliminary data, identify limitations or inconsistencies, and refine my method to improve reliability and reduce uncertainty.**
National Identity Domain مجالات الربط بالهوية الوطنية	Belonging الإنتماء Global Understanding العال م التفاهم History التاري••••خ **Area:** UAE Transportation, Road Safety, Innovation and Sustainable Infrastructure المجال Implementation: Students analyse Abu Dhabi transport motion using graphs, gradients, signed areas and kinematics equations, connecting accurate modelling to UAE road safety and smart mobility. التطبيق